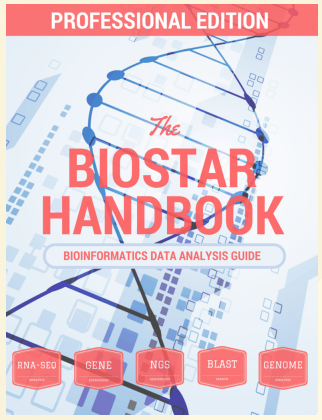
Lecture 1

Introduction to Applied Bioinformatics

Course Materials



- Biostar Handbook
- http://www.biostarhandbook.com
- Learn Bioinformatics in 100 hours



Rationale the Course

- Life sciences have become a **data driven** science.
- Most data are represented as **text based** files in various formats and need to be processed one step at a time.
- Most bioinformatics classes/books focus on algorithms and implementation details.
- This course focuses on **information processing** and **data analysis**.
- We'll teach you **what to do** with the data to extract the **information** contained inside it.

Course Requirements

A computer and a "can-do" attitude 😄

You may use `MacOS`, `Linux` or `Windows 10`

This is a good time to tell your advisor that this course recommends that you get a Mac.

Tell them I guarantee that a new Macbook Pro will make YOU incredibly productive.

Course Structure

15 weeks about two lectures per week (this may vary slightly).

- Core informatics competency.
- Computational foundations
- Biological data formats
- Software tools and their applications.
- Increasingly complex data analysis tasks.
- By the end: "publication quality" analysis.

Lecture format

Study small well defined concepts at a time.

- In each Leacture we try to cover one topic.
- Background information.
- Practical examples that tie in with the topi.c
- Class exercises + homework.

Assignments build on the lecture - **redo** what I did in class.

Complexity vs Decision Making

Bioinformatics analyses require a *very large* number of *very simple* decisions.

Most of which need to be correct! That's what makes it difficult! The software is already written for you. How you *choose* to order the steps can make a lot of difference.

There are no *absolute rules*, only guidelines. You must learn to improvise and adapt.

That what this course is about: *How to not be afraid of making decisions.*

Expectations

You can learn only by doing it!

Spend a few hours on each lecture beyond what we show:

- Explore behaviors.
- Expand the scope of the study.
- Try new solutions, push the boundaries.

Time flies when you know what you are doing.

Why do we want to study the genome?

Think about this in a brief brain storming session

- What is a genome?
- Why do we want to study it?
- What can we explain via a genome?
- What can't be answered by studying a genome?
- What complications might we run into?

Is the the Genome the Book of Life?

> *Once you “read it” you understand "everything". We just don't know how to read it yet.*

Do you consider the above true?

YES	NO

Bioinformatics Today

Combination of different sciences

1. **Information technology:** data storage, transfer, data transformation
2. **Computer science:** algorithms advanced data structures, software tools
3. **Statistics** yet traditional statistics is not well suited for modeling systematic errors over large number of observations
4. **Life Sciences:** biological hypothesis testing, interpretation

How is bioinformatics practiced?

1. **Command line tools.** "Action words" are chained together to form a pipeline. Data “flows” from one command to the other.
2. **R Programming environment.** A high level programming statistical environment. Best suited for later stages of analyes.
3. **Tools with Graphical user interfaces**: Web based interfaces to command line tools (Galaxy), large selection of commercial software.

What are command line tools like?

"Action words" that eventually become familiar:

```
bwa mem read1.fq read2fq | samtools sort > alignment.bam
```

"Chained" with characters such as `|` and `>` to form a pipeline. Data **flows** from one command into the other.

Resembles natural language. Provides generic building blocks. Adaptive and expressive. Easy to repeat or share the same command.

There is a learning curve to it.

R Programming environment

Specifically the Bioconductor package in R.

A high level statistical programming environment.

Attempts to provide with “simple” constructs that perform complex tasks. Excellent visualization capabilities

Unfortunately as a programming language R is not well designed. It is quite challenging (maddening?) to use it correctly.

What does an analysis look like in R

Provides more "specialized" action words:

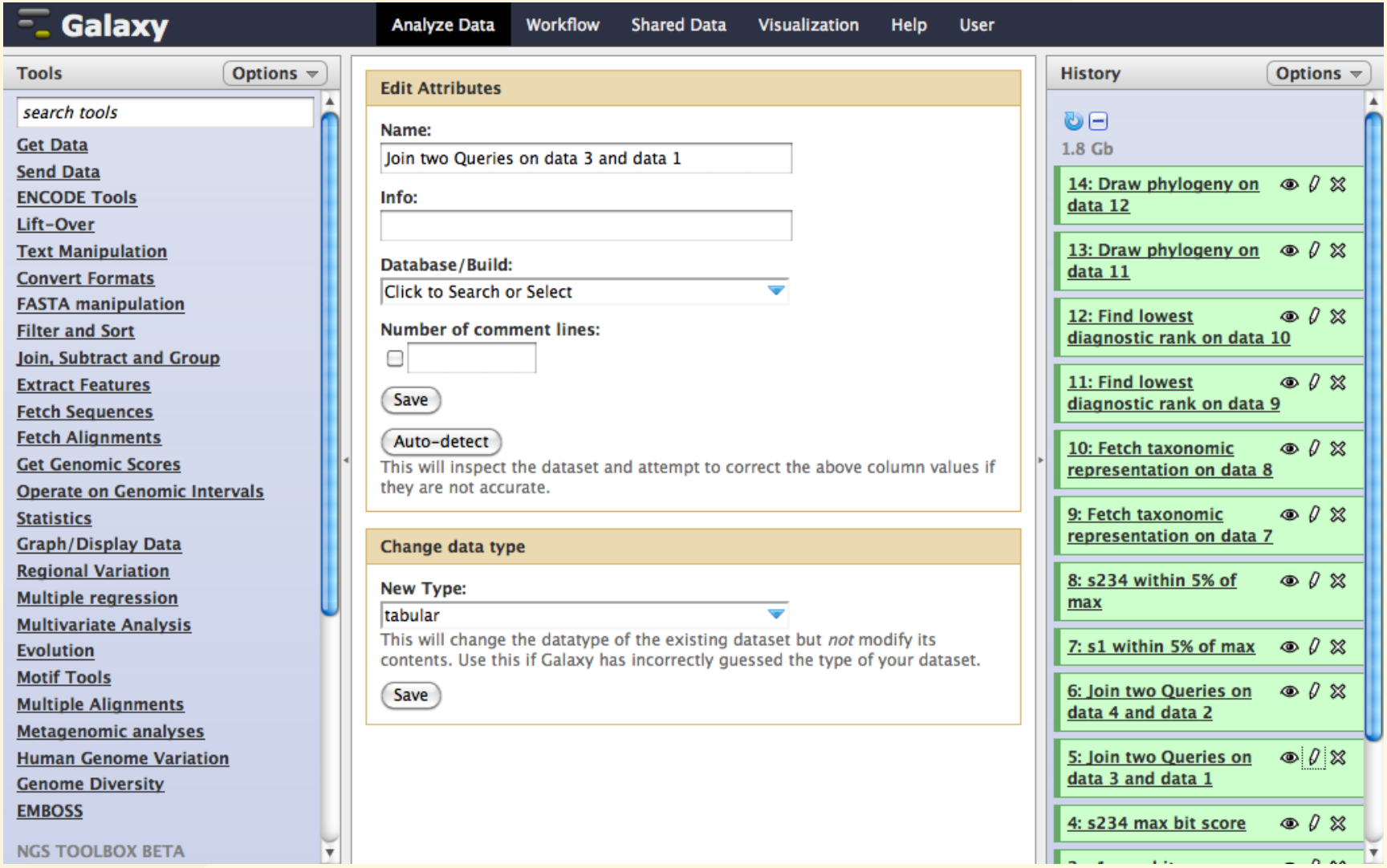
```
biocLite("DESeq")
library(DESeq)
count = read.table("stdin", header=TRUE, row.names=1 )
cond1 = c("control", "control", "control")
cond2 = c("treatment", "treatment", "treatment")
conds = factor(c(cond1, cond2))
cdata = newCountDataSet(count, conds)
esize = estimateSizeFactors(cdata)
edisp = estimateDispersions(esize)
rdata = nbinomTest(edisp, "control", "treatment")
```

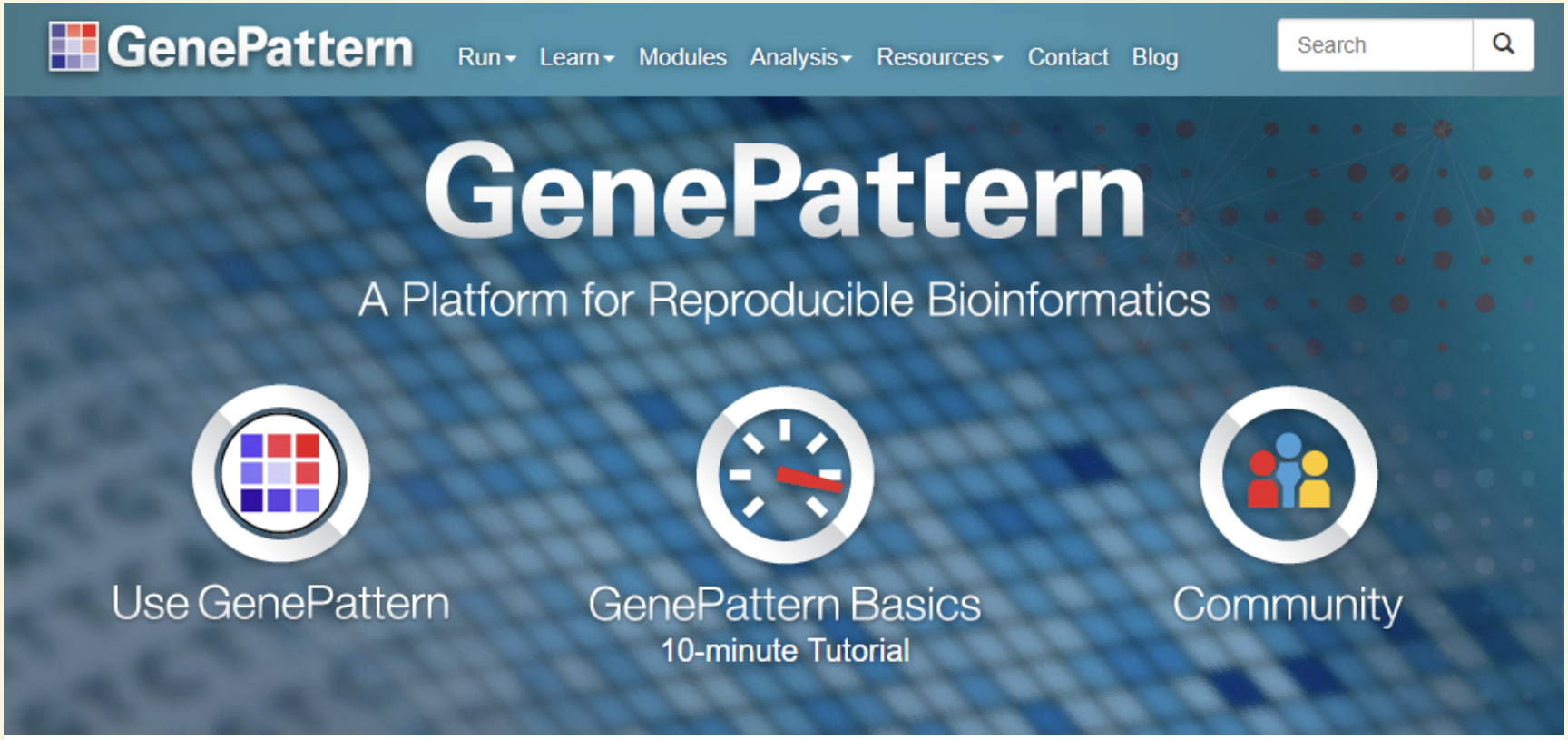
Bad: requires lots of "book-keeping". Exceedingly **easy to make mistakes** (mix up labels etc.) that could be very hard to notice.

How do grapical user interfaces work?

- So called "discoverable" environment that initially appears to be simple. This "simplicity" is deceiving.
- Often lag far behind when it comes to applying the newest of analyses.
- Difficult to understand what has been done to a given dataset.
- *"I have analyzed my data with Galaxy"* what does that even mean?
- Difficult to repeat the process the same way.

What does the Galaxy interface look like?





Features

Powerful genomics tools in a user-friendly interface

GenePattern provides hundreds of analytical tools for the analysis of gene expression (RNA-seq and microarray), sequence variation and copy number, proteomic, flow cytometry, and network analysis. These tools are all available through a Web interface with no programming experience required.

Blog > GP updates

- Scheduled Maintenance Downtime - Saturday, July 22, 2017
- End of support for GISTICPreprocess
- End of support for GENE-E
- GenePattern 3.9.10 is released!
- GenePattern Notebook Repository Update Released - v17.05

view more >

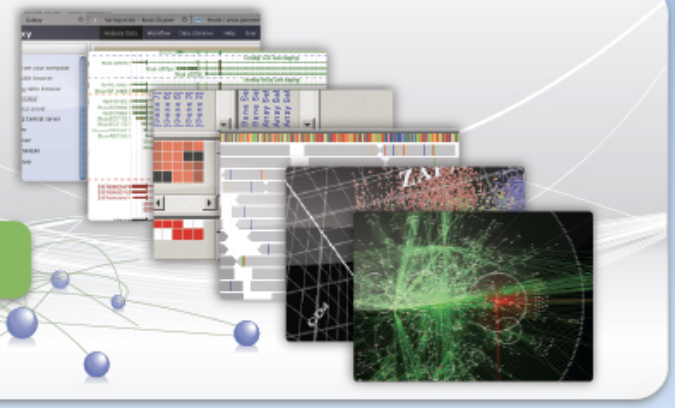
GENOMESPACE

User Login

What is GenomeSpace? Tools Recipes Documentation Developers Support About

GENOMESPACE

Frictionless connection of bioinformatics tools



Register User Login

STATUS

04.11.17 09:09PM Currently all systems are operating normally.

Calendar of Upcoming Events

Citing GenomeSpace

To cite your use of GenomeSpace, please reference Qu K, Garamszegi S, Wu F, et al. Nature Methods. 2016 Jan 18. doi: 10.1038/nmeth.3732.

F1000 Research

Check out our F1000 GenomeSpace Channel for published, community-contributed recipes.

WHAT'S NEW

News Highlights | GenomeSpace Blog

@genomespace | Mentions

Wikipedia Chain
@wikipediachain

OCLC > Public domain > List of open-source bioinformatics software > Genomespace > Computing platform > PlayStation > Tiny Pop > QVC Beauty

Aug 13, 2017

Vera Matser

What do most "graphical" tools do behind the scenes?

"Graphical" tools execute command line tools.

Wouldn't it be a lot simpler if you ran your own commands directly?

Are Bioinformatics analyses "procedural tasks"?

A procedural task involves performing a procedure, which is a sequence of activities to achieve a goal.

- Step 1,
- Step 2,
- Step 3.

Is that how bioinformatics works?

Think about the correct "task-level"

It is essential that we identify the activities at the correct level. Compare the two "tasks"

Run the `bowtie` aligner, then run the `cuffdiff` software.

Run a splice aware aligner, then quantify the abundances from an alignment file.

Tools change. Concepts don't. Over time tools implement the same concepts better and better.

Where to go next?

1. Read the additional materials.
2. Set up your computer.

Installation may pose a few unexpected challenges - system updates, passwords, downloading lots of files etc.

Get to it right away as it may take you a while to set everything up.

💻 The Biostar Handbook has detailed instructions on setting up your computer.

⚠️ Set up your computer first

You won't be able to complete assignments otherwise